ECONOMIC DATA WAREHOUSE ANALYTICS REPORT

Macroeconomic Stability & Risk Intelligence Across Emerging Economies (2005–2023)

---

1. Executive Summary

This project builds a **multi-source economic data warehouse** integrating macroeconomic indicators from the **World Bank and IMF** into a unified analytics-ready dataset.

The system transforms raw economic indicators into a structured model containing:

- GDP growth
- Inflation
- Unemployment
- Current account balance
- Government debt
- Composite economic health metrics:
    - Economic Health Score
    - Stability Score
    - Risk Classification

Key Outcome

The analysis reveals persistent structural vulnerabilities across emerging economies (Brazil, India, Kenya, Nigeria, South Africa), characterized by:

- High macroeconomic volatility
- Chronic fiscal and external imbalances
- Elevated unemployment pressure (especially South Africa)
- Frequent inflation shocks (notably Kenya and Nigeria)

Despite intermittent growth periods, **all economies remain classified as HIGH RISK over the full period**, indicating structural rather than cyclical instability.

---

2. Project Objectives

Primary Objective

To design a **data warehouse pipeline (Bronze - Silver - Gold - Analytics)** that consolidates global macroeconomic indicators into a unified analytical system for risk and performance assessment.

Specific Objectives

- Integrate multi-source economic datasets (World Bank + IMF)
- Standardize indicators across countries and years (2005–2023)
- Build a unified analytics layer for time-series economic comparison
- Develop interpretable economic health and stability scores
- Generate exportable datasets for visualization and reporting

---

3. Data Architecture Overview

Bronze Layer (Raw Data)

- Separate tables per indicator:
    - GDP growth
    - Inflation
    - Unemployment
    - Current account
    - Government debt (IMF)
- Wide format (year columns 2005–2024)
- No transformation except ingestion

Silver Layer (Standardized Model)

- Long format transformation:

country | year | indicator | value

Gold Layer (Business Logic Layer)

- Aggregated economic metrics:
    - Economic Health Score

    - Stability Score
    - Risk Category
    - Binary flags (strong economy / stress)

Analytics Layer

- Final analytical model:

country-year macroeconomic panel dataset

- Used for:
    - Trend analysis
    - Risk profiling
    - Visualization dashboards
    - CSV exports

---

4. Key Analytical Dimensions

4.1 Economic Health Score

Composite index derived from:

- GDP growth (+)
- Inflation stability (−)
- Unemployment (−)
- External balance (current account)
- Debt burden (−)

Interpretation:

- Higher score = stronger macroeconomic fundamentals
- Lower score = structural fragility

4.2 Stability Score

Measures **volatility and consistency** of macro indicators.

- Penalizes:
    - Inflation spikes
    - GDP contractions
    - Debt accumulation

Used to identify macroeconomic resilience.

4.3 Risk Classification

Category	Meaning
LOW RISK	Stable macro fundamentals
MODERATE RISK	Mixed signals
HIGH RISK	Structural instability

Result:

All countries in dataset are classified as HIGH RISK across most years, indicating systemic vulnerabilities in emerging economies.

---

5. Key Insights (Country-Level)

BR Brazil

Observations:

- Strong GDP growth pre-2010 (peaking ~7.5%)
- Severe recession in 2015–2016
- Rising government debt after 2010
- Inflation volatility post-2015

Insight:

Brazil exhibits a **cyclical economy with structural fiscal pressure**, especially post-commodity boom.

Visualization ideas:

- GDP growth vs debt line chart
- Inflation volatility heatmap
- Economic health score trend

IN India

Observations:

- Strong long-term growth (6–9%)
- Inflation spikes (2010–2013)
- Improved stability post-2015
- COVID shock in 2020 followed by rapid recovery

Insight:

India shows **growth resilience with inflation sensitivity**, but improving macro stability over time.

Visualization ideas:

- GDP vs inflation dual-axis chart
- Recovery post-2020 bar chart
- Stability score trend line

KE Kenya

Observations:

- High inflation volatility (peak ~26% in 2009)
- Moderate GDP growth (3–6%)
- Persistent fiscal deficits
- Low unemployment data stability issues

Insight:

Kenya demonstrates **inflation-driven instability and external imbalance risk**.

Visualization ideas:

- Inflation spike timeline
- Current account deficit trend
- Risk score heatmap

NG Nigeria

Observations:

- Oil-driven volatility in GDP and external balance
- High inflation (peaking ~18%)
- Weak unemployment stability
- Frequent macro shocks

Insight:

Nigeria is a **resource-dependent economy with structural inflation and fiscal instability**.

Visualization ideas:

- Oil shock period overlays
- Inflation vs GDP divergence
- Debt vs current account balance

ZA South Africa

Observations:

- Persistent high unemployment (22% - 34%)
- Low GDP growth (<3% most years)
- Rising debt and fiscal strain
- Chronic economic stagnation

Insight:

South Africa shows **structural unemployment crisis and low growth trap**.

Visualization ideas:

- Unemployment trend chart (key visual)
- GDP stagnation line chart
- Debt accumulation curve

---

6. Cross-Country Comparative Insights

6.1 GDP Growth Comparison

- India: strongest sustained growth
- Brazil & Nigeria: volatile cyclical growth
- South Africa: weakest growth trajectory

6.2 Inflation Volatility Ranking

1. Kenya (highest volatility)
2. Nigeria
3. Brazil
4. India
5. South Africa

6.3 Unemployment Pressure

- South Africa >> highest structural unemployment
- Kenya & Nigeria moderate
- India relatively stable low unemployment (historically)

6.4 Debt Burden Trend

- Rising across all countries post-2010

- Brazil and South Africa show strongest fiscal pressure

---

7. Suggested Dashboard Visualizations

You can build these in Power BI / Tableau / Streamlit:

Macro Overview Dashboard

- GDP growth trend (all countries)
- Inflation comparison line chart
- Unemployment heatmap

Risk Intelligence Dashboard

- Risk category distribution
- Economic health score ranking
- Stability score trends

Country Deep Dive Page

For each country:

- GDP vs inflation dual-axis
- Debt vs current account
- Risk score trend over time

Volatility Dashboard

- Standard deviation of GDP growth
- Inflation volatility index
- Economic shock detection (2008, 2020)

---

8. Key Findings

1. Structural Risk Dominates

All countries show persistent HIGH-RISK classification - not cyclical but structural.

2. Inflation is the Primary Instability Driver

Especially in Kenya and Nigeria.

3. South Africa is a Labor Market Crisis Economy

Unemployment is the strongest negative structural indicator.

4. India is the Most Resilient Economy

Strong growth offsets inflation and debt pressures.

5. Post-2010 Debt Expansion is Universal

All economies show fiscal expansion pressures.

---

9. Recommendations

Policy-Level Insights

- Strengthen inflation targeting frameworks (Kenya, Nigeria)
- Diversify fiscal revenue bases (Brazil, Nigeria)
- Structural labor reforms (South Africa)
- Improve external balance management (all countries)
- Sustain productivity-driven growth (India model)

---

10. Data Engineering Value

This project demonstrates:

Data Engineering

- Bronze - Silver - Gold architecture
- Cross-source integration (World Bank + IMF)
- Wide-to-long transformations

Analytics Engineering

- KPI design (health score, stability score)
- Risk modeling layer
- Time-series analytics

BI Readiness

- Fully exportable datasets
- Dashboard-ready structure
- Interpretable metrics

---

11. Conclusion

This project successfully builds a **macro-financial intelligence system** that transforms raw global economic data into a structured analytical warehouse.

It enables:

- Country risk profiling
- Economic trend monitoring
- Policy-level insights
- Comparative macroeconomic analysis